CITE EVENT – I

Problem Statement 1: Smart College Infrastructure Ticketing & Resolution System

Problem Context

Educational institutions frequently experience infrastructure, technical, and administrative issues across classrooms, laboratories, auditoriums, hostels, transport systems, and other campus facilities.

Common challenges include:

Informal or delayed issue reporting, No structured tracking mechanism, Lack of accountability ,Poor coordination between departments ,No defined resolution timeline ,Extended downtime affecting teaching and operations

To address these challenges, there is a need for a centralized, role-based Smart Ticketing System that ensures structured issue reporting, automated routing, transparent monitoring, and time-bound resolution.

Understanding SLA (Service Level Agreement)

A Service Level Agreement (SLA) is a predefined time commitment within which a reported issue must be resolved.

Why SLA is Important?

- Prevents unnecessary delays
- Defines accountability
- Ensures priority-based resolution
- Improves operational efficiency
- Enables performance measurement of departments

◆ Example:

- Critical Issue (e.g., Network down during exam) → Resolve within 2 hours
- Electrical fault in classroom → Resolve within 6 hours
- Minor maintenance request → Resolve within 24 hours

The system must track SLA automatically and trigger escalation if resolution time exceeds the defined limit.

Unified Problem Statement

Design and develop a Smart College Infrastructure Ticketing & Resolution System that enables authorized users to raise service tickets with photo/video proof and ensures:

Automated routing , Priority-based handling , SLA-based time-bound resolution ,Real-time tracking ,Verification workflow ,Escalation mechanism ,Performance analytics

MECHANISM – 1

Department-Level Smart Technical Ticketing System

Scope

- Classrooms
- Laboratories
- Auditoriums
- Seminar Halls
- Academic Common Areas

Service Teams

- IT / System Administration
- Electrical
- Civil Maintenance
- Network
- Audio-Visual
- Housekeeping

Workflow

Faculty/Lab Assistant raises ticket

System auto-routes to relevant technical team

3SLA timer starts automatically

Technician updates status

Faculty verifies and closes

Key Requirements

Role-based login (Faculty, HOD, Technician, Admin)

Ticket creation with:

Location (Block / Room / Lab)

Issue category

Description

Photo/Video upload

Automated routing to service desk

Priority levels:

- Critical
- High
- Medium
- Low

Real-time status tracking:

- Open → In Progress → Resolved → Verified → Closed
- SLA countdown timer display
- Escalation if SLA breached
- Feedback and rating system

Expected Outcome (Mechanism 1)

- Reduced academic downtime
- Faster technical resolution
- Clear technician accountability
- SLA-driven performance tracking

MECHANISM – 2

Integrated AO-Based Campus Infrastructure & Administrative System

Extended Scope

- Includes all academic areas plus:
- Hostel
- Quarters
- Security
- Gardening
- Transport & Drivers
- HR
- Store
- Visitor Management
- Hall Booking System
- All governed under the Administrative Office (AO).

Governance Structure

- Requester – Raises ticket
- Department Head – Assigns & defines SLA
- Execution Team – Resolves & uploads proof
- Admin Operations – Monitors escalations
- AO – Strategic oversight & analytics

Extended Workflow

Ticket Creation

→ Auto-forward to Department Head

→ SLA defined during assignment

→ SLA timer activated

→ Execution with status updates

→ Completion proof upload

→ Requester verification

→ Closure / Reopen

→ Escalation up to AO if SLA exceeded

SLA Escalation Hierarchy

- If resolution exceeds SLA:
- Level 1 → Reminder to Technician
- Level 2 → Alert to Department Head
- Level 3 → Escalation to Admin Operations
- Final Level → Escalation to AO

Dashboard Requirements

AO Dashboard

- Sector-wise ticket overview
- SLA compliance percentage
- Overdue ticket alerts
- Department performance comparison
- Monthly analytics

Department Dashboard

Assigned tickets

SLA countdown

Pending / Completed

Reopened tickets

Reports & Data Management

- Excel export
- Filter by status, priority, department, date
- Secure media storage
- Maintenance history tracking
- SLA performance reports

Advanced Features (Optional)

- AI-based issue classification
- Predictive maintenance alerts
- QR-based room/asset scanning
- Mobile app interface
- Chatbot ticket creation
- Analytics-driven workload balancing

Final Expected Outcome

- The proposed system should deliver:
- Centralized and transparent service management
- SLA-based time-bound resolution
- Reduced downtime across campus
- Improved accountability
- Data-driven administrative decisions
- Scalable enterprise-ready architecture